ZCAS UNIVERSITY

SCHOOL OF COMPUTING, TECHNOLOGY & APPLIED SCIENCES

CAS5821 MATHEMATICAL FOUNDATIONS OF DATA SCIENCE

DEFERRED EXAMINATION

DATE: 18th June 2026

TIME:14:00hrs – 17:00hrs

TIME ALLOWED :WRITING – THREE HOURS

READING – 5 MINUTES

INSTRUCTIONS:

1. Section A: This question is **compulsory** and must be attempted.
2. Sections B: Answer **TWO (2)** questions from this section.
3. This examination paper carries a total of **100 marks**.
4. Candidates must **not turn this page** until the invigilator tells them to do so.

SECTION A: Question 1 is compulsory and must be attempted.

Question 1

Every summer there is a plea by the Water Authority for the public to use water wisely. The Authority suspects that water use is related to the temperature on the day. To establish this, it collects the following data on water use and the day's temperature.

Water Use (million gal.)	**Temperature** (Fahrenheit)
219	103
56	39
107	77
129	78
68	50
184	96
150	90
112	75

(i) Calculate the linear regression equation that describes the relationship between water usage and the day's temperature. [20 Marks]

(ii) Interpret the coefficients a and b. [6 Marks]

(iii) Calculate the correlation coefficient and coefficient of determination and interpret their meaning. [14 Marks]

(iv) What is the predicted water usage for a day with a temperature of 100^0. [10 Marks]

(50 marks)

SECTION B: Attempt any TWO (2) questions in this section

Question 2

External Students enrolment in 1995 is cross-classified by three course levels and their gender in the contingency table below:

Course level	Male	Female	**Total**
Masters	40	31	**71**
Postgraduate Diploma	11	12	**23**
Graduate Diploma	118	311	**429**
Total 169	**354**	**523**	

If a student is selected at random from this sample, what is the probability that:

(i) the student is enrolled in a postgraduate diploma program? [5 Marks]

(ii) the student is a male enrolled in a masters program? [5 Marks]

(iii) the student is enrolled in the masters program and the graduate diploma program? [5 Marks]

(iv) If a student selected at random is a female student, what is the probability that she is enrolled in a Masters program? Is the level of the course the student enrols in dependent on their gender? [10 Marks]

(25 marks)

Question 3

a) Give five examples where regression Analysis can be used in solving real life problems. [10 Marks]

b) Calculate the product moment correlation coefficient and hence the coefficient of determination (r^2) for the set of data given below and interpret the results:

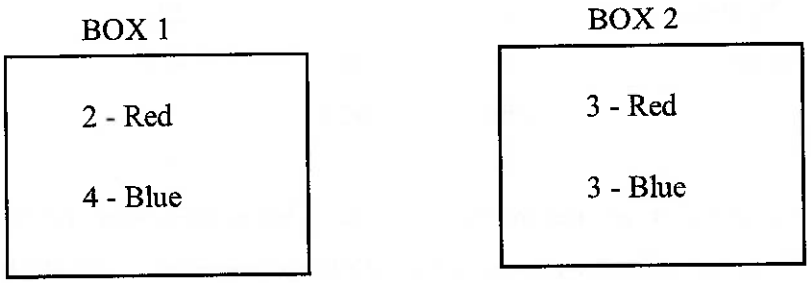
X	3	4	6	7	8	10	12	15	16
Y	3.2	2.9	3.7	2.2	1.8	2.3	1.7	0.8	0.3

[15 Marks]

(Total: 25 marks)

Question 4

a) Suppose you have two boxes labelled Box 1 and Box 2. In Box 1 there are 2 red chalks and 4 blue chalks. In Box 2 there are 3 red chalks and 3 blue chalks. The probability of picking any one of the two boxes is half. Within the boxes the probability of picking a red or a blue chalk is given below:



P(Box 1) = 1/2	P(Box 2) = 1/2
P(Red|Box 1) = 2/6	P(Red|Box 2) = 3/6
P(Blue|Box 1) = 4/6	P(Blue|Box 2) = 3/6

Let's assume that the two boxes are covered and you are asked to pick a chalk. Not knowing which box your hand is in, you pull out a red chalk.

What is the probability that the red chalk came from Box 1? [15 Marks]

b) Find $\frac{dy}{dx}$ of the following:

(i) $y = 4x^3 + 5x^2 - 1$ (ii) $y = \sqrt{2x-1}$ [10 Marks]

(25 marks)

END OF EXAMINATION